

Lucie Gaspard-Boulinc

PHD CANDIDATE IN BIOINFORMATICS · CANCER GENOMICS · SPATIAL BIOLOGY

PhD defense on September 9, 2026. Based in Paris, France

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Summary

Profile: PhD candidate specializing in bioinformatics for human pathologies. My ongoing research focuses on heterogeneities of brain tumors by harnessing spatial transcriptomics. I work at the interface of computational method application and translational research, collaborating with biologists and clinicians. I am seeking post-doctoral opportunities to contribute to a better understanding and targeting of human pathologies.

Technical expertise: Spatial and single-cell transcriptomics analysis, R/Python for bioinformatics, statistical modeling.

Education

Institut Curie - Paris Sciences et Lettres

Paris, France

PHD IN BIOINFORMATICS UNDER THE SUPERVISION OF FLORENCE CAVALLI AND EMMANUEL BARILLOT

Sept. 2022 - Now

- Title: Deciphering spatial heterogeneities of IDH-mutant gliomas
- Defense on September, 9th, 2026

École normale supérieure - Ulm

Paris, France

M.SC DEGREES WITH DIPLOMA OF THE ENS, ADMITTED AFTER NATIONAL COMPETITIVE EXAMS (TOP 2.5%)

Sept. 2018 - Sept. 2022

- M.Sc in System biology, bioinformatics and genomics: transcriptomic and epigenomics data analysis, computational biology, protein biochemistry, R/Python programming, biostatistics;
- B.Sc in Life Sciences: molecular cancer genetics, epigenetics, functional genomics, ecology, neurosciences.

Sorbonne Université

Paris, France

M.SC DEGREE IN INTEGRATIVE BIOLOGY - SPECIALIZATION IN HUMAN PATHOLOGIES

Sept. 2020 - Sept. 2021

- Cancer biology, cardiovascular diseases, lung diseases, immunology, pharmacology.

Research Experience

PhD Researcher in Cancer Bioinformatics

Paris, France

INSTITUT CURIE – COMPUTATIONAL BIOLOGY AND INTEGRATIVE GENOMICS OF CANCER TEAM

Sep. 2022 – Present

- Characterized spatial organization of IDH-mutant gliomas across 20 patients samples using Visium spatial transcriptomics, identifying recurrent cellular niches, their spatial distribution and interactions;
- Developed R/Python analysis pipelines integrating QC, deconvolution, and spatial statistics;
- Led interdisciplinary collaboration with clinicians and pathologists to validate computational findings against histopathology;
- First-author manuscript in preparation [1]; co-authored comprehensive review on deconvolution methods [2];
- Co-supervised 4 undergraduate and graduate students on bioinformatics analysis projects.

Graduate intern in computational biology

Paris, France

INSTITUT CURIE – COMPUTATIONAL BIOLOGY AND INTEGRATIVE GENOMICS OF CANCER TEAM

Feb. 2022 – Aug. 2022

- Systematic comparison of cell-type deconvolution methods for spatial transcriptomics data;
- Implemented benchmarking workflows and statistical evaluation pipelines in R.

Graduate intern in protein biochemistry for cardiovascular disease

Paris, France

INSTITUT DE MYOLOGIE – STRIATED MUSCLES AND SIGNALLING TEAM

Feb. 2021 – Jun. 2021

- Investigated ADP-ribosylation mechanisms in dilated cardiomyopathy mouse models;
- Developed experimental protocols to characterize differentially ADP-ribosylated proteins, combining biochemical assays and molecular biology approaches.

Graduate intern in single-cell genomics for blood cancer

Sydney, Australia

CHILDREN'S CANCER INSTITUTE - CANCER AND STEM CELL TEAM

Feb. 2020 – Jul. 2020

- Characterized gene regulatory networks in GADD45A-knockout AML cell lines using CITE-seq multi-modal single-cell data;
- Led computational analysis identifying dysregulated genes, contributing to publication in Blood [3] demonstrating GADD45A's role in stem cell maintenance.

Undergraduate intern in cell biology

Toulouse, France

CENTRE DE RECHERCHE EN CANCÉROLOGIE DE TOULOUSE - IMPACT

Jun. 2019 – Aug. 2019

- Assessed the effect of nucleotide pool imbalance on replicative stress in CDA overexpressed pancreatic cancer cell lines;
- Performed cell culture, flow cytometry, immunofluorescence and quantitative image analysis.

Publications

1. (In preparation) **Gaspard-Boulinc, L.C.**, et al. *Principles of spatial architecture in IDH-mutant gliomas*.

2. **Gaspard-Boulinc, L.C.**, Gortana, L. et al. *Cell-type deconvolution methods for spatial transcriptomics*. **Nature Reviews Genetics** 2025; 26, 828–846 . doi:10.1038/s41576-025-00845-y.

3. Hassan N, Yi H, Malik B, **Gaspard-Boulinc L**, et al. *Loss of the stress sensor GADD45A promotes stem cell activity and ferroptosis resistance in LGR4/HOXA9-dependent AML*. **Blood**. 2024; 144(1):84-98. doi:10.1182/blood.2024024072.

Skills

Bioinformatics & Computational Biology

- Spatial transcriptomics data analysis: Visium processing, cell-type deconvolution, spatial statistics, cell-cell communication
- Single-cell genomics: scRNA-seq/CITE-seq analysis, multi-modal integration
- Statistical modeling: hypothesis testing, spatial statistics, dimensionality reduction, linear models

Technical infrastructure

- Programming: R (Seurat, Bioconductor), Python (Scanpy/Squidpy, scverse), bash
- HPC computing (SLURM), large-scale data processing
- Environment manager: conda, renv

Biology

- Cell line culture and treatment
- Molecular biology: western blot, PCR
- Flow cytometry, protein biochemistry

Languages & communication

- English (proficient), French (Native)
- Pack Office: PowerPoint, Excel
- Inkscape, BioRender

Conference presentation

Poster presentation

2025	Society of Neuro-Oncology , Unravelling IDH-mutant heterogeneities with spatial transcriptomics	Honolulu, USA
2024	Oligodendroglioma Workshop - POLA , on oligodendroglial tumors profiled with ST - Best poster award	Paris, France
2024	VIB - Spatial Omics Conference , Deciphering spatial heterogeneity of IDH-mutant gliomas	Ghent, Belgium
2022	Computational System Biology of Cancer , on deconvolution method benchmarking - Best poster award	Paris, France

Oral presentation

2026	10X Genomics Focus Day , Cell-type deconvolution for ST: lessons from brain tumor data	Paris, France
2025	J.O.B.I.M , Cell-type deconvolution for spatial transcriptomics	Bordeaux, France

Teaching and community

Teaching experience

2022-2025	R/Python programming and biological data analysis , ENS - PSL, 192 total hours (BSc/MSc level)	Paris, France
2025	Workshop on ST data analysis , Qlife Quantitative Biology Spring School	Paris, France
2023	Single-cell genomics tutorial , PR[AI]RIE summer school - AI4Health	Paris, France

Community

2023-2025	PhD student and Post-Doc representative , U1331 - Computational Oncology Department	Paris, France
2022-2024	Working group on spatial transcriptomics , Coordinated the serie of 10 seminars on ST	Paris, France

Science communication

2025	My Thesis in 180 seconds , PSL finalist - "Les cancers du cerveau ce n'est pas de la tarte"	Paris, France
2025	Pint Of Science , "Brain tumors: a space odyssey"	Paris, France

References

Florence Cavalli Principal Investigator PhD main supervisor florence.cavalli@curie.fr +33 1 47 11 17 32	Emmanuel Barillot Research Director PhD co-supervisor emmanuel.barillot@curie.fr +33 1 56 24 69 79	Nunki Hassan Research fellow Master thesis supervisor nunki.hassan@sydney.edu.au	Morgane Thomas-Chollier Assistant Professor Teaching supervisor morgane.thomas-chollier@france-bioinformatique.fr
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